IHX Nucleus Integration Design

Overview

This document entails information regarding the API Request and Response attributes for seamless integration and efficient data retrieval. The response attributes provide detailed insights into the data returned by API Request including key parameters, required and optional fields and data formats.

The envisioned integration operates as follows: Whenever a hospital insurance desk user accesses IHX and searches for a patient using their unique identifier (such as name, mobile number, IP number, UHID, etc.), the system retrieves all pertinent information from the Hospital Information System (HIS), including patient demographics, treatment specifics, billing details, and relevant claim documents. In this process, the hospital's insurance desk user acts as the verifier of the information and proceeds to submit the claim. By eliminating manual processes and paper documentation, this integration streamlines claim submission, reducing processing time and minimising input errors, ultimately enhancing overall claim processing efficiency.

Assumptions about the Hospital Information System (HIS)

This document operates under certain assumptions regarding the functionality and structure of the Hospital Information System (HIS) in question. These assumptions are essential for understanding the context and scope of the information presented.

- **Unified Data Storage**: It is assumed that all relevant data and documents pertinent to the HIS are centralised within a single, cohesive system.
- **Comprehensive Status Tracking**: The HIS is designed to capture and manage all relevant statuses at every stage of the process. Any change in status is considered integral to the system's functionality.
- **Notification of Changes to the Schema**: It is presumed that the HIS will promptly notify IHX of any modifications to data formats or field structures. This allows for proactive adjustments in the integration and includes deprecations, revisions to existing data, and new fields.
- **Real-time Patient Data Retrieval:** The assumption is made that when conducting a patient search using the in-patient number, the HIS retrieves the latest available data. This ensures that the information presented is current and may differ from previously retrieved data during the patient's hospital stay
- **Structured Billing Information:** While PDF forms are utilised for billing, it is assumed that detailed itemised billing information is also available in a digital format, such as JSON. This structured form of billing data is presumed to be accessible for consumption and analysis.
- **Interoperability:** The HIS is assumed to have the capability for interoperability with other systems or applications, allowing seamless data exchange and integration with external healthcare platforms.
- **Standardised codes**: It is presumed that the HIS adheres to industry standards like ICD-10 and CPT and employs standardised code sets for medical diagnoses, treatments, and other pertinent healthcare data.
- **Uniform Data Format**: It is assumed that there is a standard format for the data transferred between the HIS and IHX, which facilitates processing and lowers the possibility of data mapping mistakes.
- **Notification Mechanisms:** The HIS is assumed to have notification mechanisms in place to alert relevant stakeholders about critical events, such as claim rejections or updates to patient records.
- **Handling Field Updates**: It is anticipated that a versioning technique will be used if any system modifications impact fields in the HIS or IHX platform. This guarantees that earlier iterations of the fields and related data are preserved and not changed, facilitating the study of historical data and upholding adherence to audit specifications.

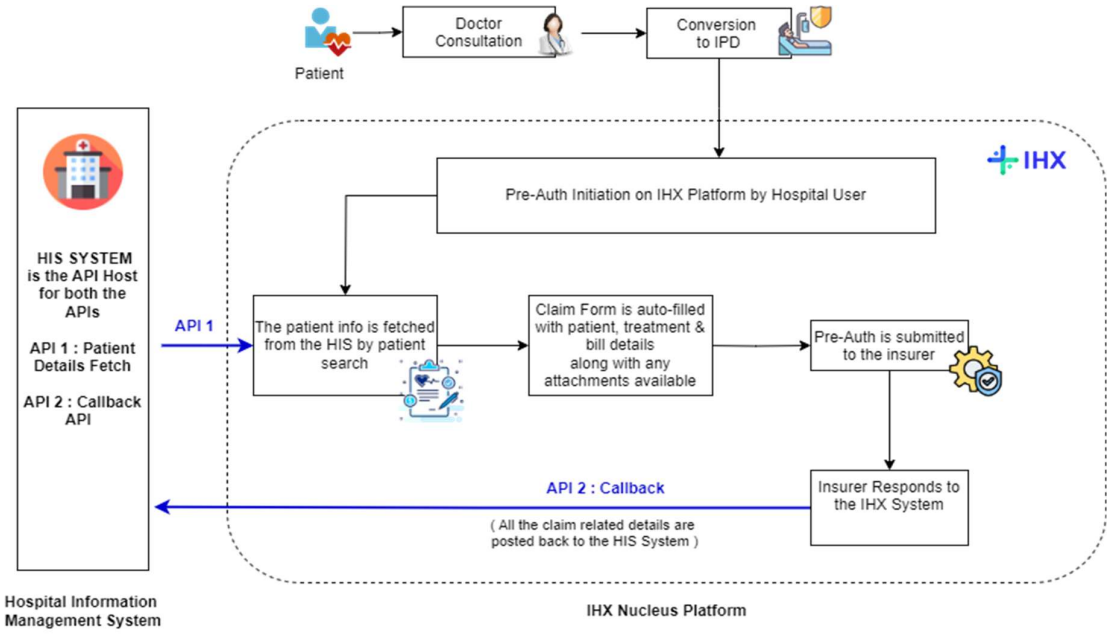
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- **Testing Environment Updates**: The HIS is assumed to maintain an updated testing environment that mirrors the production environment, allowing external clients to validate their integration against the latest system updates before they are deployed to the production environment.

S. No	Use Case	Data Attributes	Data Flow	API Host	API #
IHX001	Get New Admission (Pre-Auth) data from HIS	• Patient Demographics • Treatment Details • Bill Details (IRDAI L1 Classification)	HIS → IHX	HIS	API 1
IHX002	Get Enhancement data from HIS	• Patient Demographics • Treatment Details • Bill Details (IRDAI L1 Classification)	HIS → IHX	HIS	
IHX003	Get Discharge data from HIS	• Patient Demographics • Treatment Details • Bill Details (IRDAI L1 Classification)	HIS → IHX	HIS	
IHX004	Get Query Response data from HIS	• Patient Demographics • Treatment Details • Bill Details (IRDAI L1 Classification)	HIS → IHX	HIS	
IHX005	Get Settlement data from HIS	• Patient Demographics • Treatment Details • Bill Details (IRDAI L1 Classification)	HIS → IHX	HIS	
IHX006	Fetch Documents from HIS	All the patient documents like Bills, Reports, Discharge summary, KYC,etc.	HIS → IHX	HIS	API 2
IHX007	Send case data to HIS	• Claim Details • Approval / Denial Information • Letters	IHX → HIS	HIS	API 3

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Workflow



1. Patient Arrival at the Hospital

- The patient enters the hospital *#PatientArrival*
- The patient registers at the reception
- Receives a UHID (Unique Health Identification) number

2. OPD Consultation

- The patient goes for a doctor consultation in OPD (Outpatient Department) *#OPDConsultation*
- Consultation with the doctor
- Medical tests if required
- Prescription and follow-up instructions

2.1. Conversion to IPD

- If the patient needs further treatment, convert to IPD (Inpatient Department) *#IPDConversion*
- Admission procedures
    - Room allocation
    - Treatment plan discussion
- Or an Estimation of the procedure

3. Cashless Claim Initiation on IHX Website (First Touchpint)

- Initiate a cashless claim pre-auth on the IHX (Insurance Health Exchange) website #CashlessClaim
- Search for patient details using any one of the parameters:
    - Patient name
    - IP number
    - UHID
    - Mobile number
    - ABHA address or number

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3.1. API Integration with HIS System

- API host should be the HIS (Hospital Information System) #APIIntegration
    - Patient information flows from HIS to IHX
    - All the available data from the HIS is auto filled in the claim form .i.e info related to a particular patient's like member details, treatment details,bill details and any available documents
    - Submit the claim to the insurer on IHX

4. Insurer Response

- Handle responses from the insurer #InsurerResponse
    - Claim number
    - Claim status
    - Any queries
    - Claimed/Approved/Settled amount
    - Any Letters (Approval/Denial/Query)

4.1. API Integration for Response Handling

- Response handling is also hosted by HIS system #APIIntegration
    - Any response from the insurer passed on to the HIS from IHX via API

Multiple Visits Scenario

When a patient has multiple visits, the IHX platform can display all the available visits from the HIS as visit cards once the patient search is successful using the GetPatientDetails API. These visit cards will show minimal information, allowing the user to select the specific visit or treatment for which the pre-authorization needs to be raised . Basis the Visit TYpe & Visit Code Identifiers

Technical Aspect

➢ **API Host**: This refers to the system that initiates the API request. In all cases, the IHX is the one requesting HIS.
    - HIS → IHX: Indicates the direction of data flow. Data travels from the Hospital Information System to the IHX in all the mentioned use cases.
    - IHX → HIS: In the case of IHX007, data is sent from IHX to the Hospital Information System.

➢ **Authentication**: To authenticate your requests, include your AccessToken in the headers of HTTP requests. The AccessToken should be passed in the Authorization header as follows:

```
curl --location --request POST 'https://<HospitalIntegrationSystem>/api/PatientSearch' \
--header 'Content-Type: application/json' \
--header 'AccessToken: 5d4ad473ff3601aa4e1444fa6f47ebb809978d' \
"Source": "IHX",
"PatientID": "",
"MobileNo": "xxxxxxxxx09"
}'
```

➢ **Endpoint URLs**: For Example
    - Endpoint 1, 2 & 4: This API searches Patient details with Patient's mobile number
        - URL: https://<HospitalIntegrationSystem>/api/PatientSearch
        - HTTP Method: [POST]

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- Endpoint 3: This API is to update the claim status in HIS
  - URL: https://<HospitalIntegrationSystem>/api/PatientStatusUpdate
  - HTTP Method: [POST]

➢ **Hospital Identification**
- Hospital identification will be done by the "ProviderID" field sent as part of the request body.
- These hospital codes are mutually understood by the HIS and IHX.

➢ **Error Codes**: Throughout the API communication between the two systems, these error codes will be employed to specify the type of error that occurred during processing.
- E101: AccessToken does not exist in the API header
- E102: No key to search
- E500: Internal Error

Request & Response Parameters - API 1

The list of IHX fields required for integration and claim submission to cover use case IHX001 to IHX005 namely
- New Admission
- Enhancement
- Discharge
- Query
- Settlement

Request Parameters	HIS Integration (Required / Optional)	Claim Submission (Required / Optional)	Notes
Hospital ID	Required	NA	IHX Identifier (Master Data)
UHID	Any one of these parameters is Required	NA	Free Text Limit: 25 Characters
IP Number		NA	Free Text
Mobile Number		NA	Numeric - 10 Digit
ABHA Address (UHI)		NA	Alpha-Numeric - address@abdm
ABHA Number		NA	Numeric - 14 Digit

Response Parameters	HIS Integration (Required / Optional)	Claim Submission /NHCX (Required / Optional)	Notes
Integration Fields			
Hospital ID	Required	Required	IHX Identifier
Patient Details			
Patient Name	Optional	Required	Free Text Limit: 30 characters

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Patient Age	Optional	Optional	Numeric Limit : 3 characters
PatientGender	Optional	Optional	Male/Female/Others
Mobile Number	Optional	Required	Numeric - 10 Digit
Member ID	Optional	Optional	Alpha-Numeric Limit: 30 characters
Client ID (Same as Member ID)	Optional	Optional	Data Type: **String** Maximum Field Length: **30 characters**
Email ID	Optional	Optional	Free Text - Email Format Limit: 30 characters
ABHA ID	Optional	Optional	Numeric - 14 Digit
ABHA Address	Optional	Optional	
Employee ID	Optional	Optional	Free Text Limit: 30 characters
Insurer Name	Optional	Required	IHX Insurer / Payer Master
Payer/TPA	Optional	Required	IHX Insurer / Payer Master
Corporate Name	Optional	Optional	Free Text Limit: 30 characters
Policy Type	Optional	Optional	Retail/Corporate
Policy Number	Optional	Optional	Alpha-Numeric Limit: 30 characters
Certificate Number (COI)	Optional	Optional	Alpha-Numeric Limit: 30 characters
Inpatient Number	Optional	Optional	Alpha-Numeric Limit: 30 characters
Registration Number/UHID	Optional	Optional	Alpha-Numeric Limit: 25 characters
PAN Card Number	Optional	Optional	Alpha-Numeric
Treatment Details			
Date Of Admission	Optional	Required	Epoch IST
Date of Discharge	Optional	Required	Epoch IST
Date Of Surgery	Optional	Mandatory in case of maternity	Epoch IST
Proposed Line of Treatment	Optional	Required	**Expected values** - Surgical Management - Medical management - Investigation - Intensive Care

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Treatment	Optional	Required	
Diagnosis	Optional	Required	Option - 1: ICD-10 mappable diagnosis text Option - 2: ICD-10 code of diagnosis
Treatment Details - Doctor Details			
Doctor Name	Optional	Required	Free Text Limit: 30 characters
Doctor Registration No.	Optional	Optional	Free Text - Alpha-Numeric Limit: 30 characters
Doctor Specialisation	Optional	Optional	Free Text Limit: 30 characters
Room type	Optional	Required	Room Type Master from HIS
Is the patient getting discharged today?	Optional	Optional	- Yes - No
Is it a death case?	Optional	Optional	- Yes - No
Treatment Details - Past History			
Past Ailments - Name	Optional	Optional	**Ailment Master** - Osteoarthritis - Hypertension - Hyperlipidemia - Heart Disease - Diabetes - Cancer - Asthma/COPD/Bronchitis - Any HIV or STD/related ailments - Alcohol or drug abuse
Past Ailment Duration	Optional	Optional	Numeric - Duration in Years
Other ailments	Optional	Optional	Free Text - Ailment Name Limit : No limit
Admission Type	Optional	Optional	- Planned - Emergency
Case Details - Road Traffic Accident			
Date of Injury	Optional	Optional	Epoch Date
FIR No.	Optional	Optional	Free Text - Alpha-Numeric Limit: 30 characters
Is it an RTA?	Optional	Optional	- Yes - No
Injury or disease caused due to substance	Optional	Optional	- Yes - No

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abuse/alcohol consumption?			
Reported to Police	Optional	Optional	- Yes - No
Test conducted to establish this	Optional	Optional	- Yes - No
Bill Details			
Package	Optional	Required (if Package)	Numeric
Claimed Amount	Optional	Optional	Numeric
Room rent and nursing charges	Optional	Required	Numeric
Consultation	Optional	Optional	Numeric
OT charges	Optional	Optional	Numeric
Surgeon & anaesthetist	Optional	Optional	Numeric
Investigation	Optional	Optional	Numeric
Medicine and consumables	Optional	Optional	Numeric
Other hospital expenses	Optional	Optional	Numeric
Documents (for Claim Submission)			
Claim Form	Optional	Required	**PDF** **PNG / JPG / JPEG** **JSON** **XLS / XLSX / CSV** **TIF / TIFF**
KYC	Optional	Required	
Consultation/Prescription	Optional	Optional	
Investigation Reports	Optional	Optional	
Discharge Initiation Details			
Date Of Admission	Optional	Required	Epoch IST
Date of Discharge	Optional	Required	Epoch IST
Has the patient utilised this PA?	Optional	Required	- Yes - No
Documents	Optional	Required	Expected Documents - Discharge Summary - Final Bill - Indoor Case Papers - All Investigation Reports - Others (Document size per file should be less than or equal to 15MB)

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Patient Paid Amount	Optional	Required	Numeric
Inpatient No.	Optional	Required	Alpha- Numeric
Receipt No.	Optional	Required	Alpha-Numeric

Document Type Master

The below table outlines the preferred formats for documents to be posted as part of the cashless claim submission process. Please ensure compliance with these specifications for smooth integration and processing

DOCTYPE ID	Name	Details	HIS Integration (Required/Optional)	Claim Submission (Required/Optional)	Expected Format	Document ID
148	Bills	Bills/Receipts	Optional	Required	**PDF PNG / JPG / JPEG JSON XLS / XLSX / CSV TIF / TIFF**	**Required for each of the documents**
147	Claim Form	Claim Form / Preauth Form	Optional	Required		
150	Consultations Papers	Consultations Papers / Prescriptions / other Documents	Optional	Required (If any)		
149	Discharge Summary	Discharge Summary /Death Summary/Transfer Certificate	Optional	Required		
152	KYC	KYC/Photo ID Proof/Address Proof	Optional	Required		
153	Others	Others	Optional	Optional		
154	QueryReply	QueryReply	Optional	Required (If any)		
151	Reports	Reports/Investigations	Optional	Required(Depends on the treatment)		

Below Callback API covers use case IHX006 namely Status Update - API 3

Callback Request Parameters	HIS Integration (Required / Optional)	Claim Submission (Required / Optional)	Notes
Hospital ID	Required	NA	IHX Identifier
UHID	Any one of the parameters is Required	NA	Free Text Limit: 25 characters
IP Number		NA	Free Text Limit: 30 characters
Mobile Number		NA	Numeric - 10 Digit
ABHA Address (UHI)		NA	Alpha-Numeric - address@abdm
ABHA Number		NA	Numeric - 14 Digit
Status ID	Optional	NA	IHX Status Master
Claim Number	Optional	NA	Alpha-Numeric
Status Name	Optional	NA	IHX Status Master

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Claimed Amount	Optional	NA	Numeric
Approved Amount	Optional	NA	Numeric
Settled Amount	Optional	NA	Numeric
UTR No.	Optional	NA	AlphaNumeric
Letters	Optional	NA	Document
Remarks (At all stages Query, approval, denial)	Optional	NA	Free Text

Callback Response

Callback Response Parameters	HIS Integration (Required / Optional)	Claim Submission (Required / Optional)	Notes
Hospital ID	Required	NA	IHX Identifier
UHID	Optional	NA	Free Text
IP Number	Optional	NA	Free Text
Mobile Number	Optional	NA	Numeric - 10 Digit
ABHA Address (UHI)	Optional	NA	Alpha-Numeric - address@abdm
ABHA Number	Optional	NA	Numeric - 14 Digit
Status Name	Required	NA	Eg: Success
Status ID	Required	NA	Eg: 200
Message	Optional	NA	Eg: Saved Successfully

Treating_Doctor_Metadata			
1	Name of Treating Doctors	Name of the doctor	String
2	Specialisation	Specialisation of the doctor	String
3	Registration_Number	Registration number of the doctor	String
Diagnosis_Details			
1	Diagnosis Name	The disease name	String
2	Standard Name	Description of the ICD 10 code	String - Optional Data
3	ICD 10	ICD 10 Code	String - Optional Data
4	SNOMED Code	SNOMED code of the Diagnosis Name	String - Optional Data

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5	Is Primary Diagnosis?	Primary diagnosis is the main diagnosis having more impact. There are two entries 1. Yes : If given diagnosis is primary 2. No : If given diagnosis is not primary (Note: There can be only one primary diagnosis)	Boolean
Conditions / Symptoms			
1	Symptom	The complaint for which patient visited the hospital	String
2	Standard Name	Description of the ICD 10 code	String - Optional Data
3	ICD 10	ICD 10 Code	String - Optional Data
4	SNOMED Code	SNOMED code of the Complaint name	String - Optional Data
5	Duration	Duration of the complaint e.g., from a week, 2days back etc.	String
6	Description	Details about the complaint. eg. If the complaint text given in discharge summary as "fever since 3 days on and off" then the complaint will be fever duration since 3 days description will be on and off	String
Course in Hospital			
1	Course in Hospital	This input will be entire text given in course in hospital	String
Physical Examination			
1	Physical Examination	This input will be entire text given in examination section (Note: examination at time of admission) 1. Vital parameters 2. Edema 3. Pallor 4. Icterus 5. Lymphadenopathy etc.	String
2	Systemic Examination - System Examined	Name of System Examined 1. Respiratory System 2. Cardiovascular System 3. Abdomen 4. Central Nervous System	String

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3	Systemic Examination	This input will be entire text given in examination section (Note: examination at time of admission)	String
4	Local Examination - Site Examined	Specific Examination related to the patients complaint	String
5	Local Examination	This input will be entire text given in examination section (Note: examination at time of admission)	String
History			
History_1_PreExisting_Condition			
1	Pre-existing Condition	The disease name	String
2	Standard Name	Description of the ICD 10 code	String - Optional Data
3	ICD 10	ICD 10 Code	String - Optional Data
4	SNOMED	SNOMED code of the Condition name	String - Optional Data
5	Duration	Since how long patient is suffering from disease	String
History_2_Menstrual_History			
1	Applicable	As menstrual history is applicable to females of reproductive age group. There are two values only 1. Yes 2. No	Boolean
2	Last Menstrual Date	Date of last menstruation	Date
3	Regularity	There are two options 1. Regular 2. Irregular	Boolean
4	Complaints	Any complaint regarding menstruation	String
History_3_Obstetric_History			
1	Applicable	Applicable only in case of pregnancy two values 1. Yes 2. No	Boolean

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2	Gravida	Abbreviated as G, (Note: number denoting total number of pregnancies)	Number
3	Para	Abbreviated as P, (Note: number denoting total number of viable pregnancies)	Number
4	Live	Abbreviated as L, (Note: number denoting total number of live children)	Number
5	Abortion	Abbreviated as A, (Note: number denoting total number of abortions)	Number
6	Death	Abbreviated as D, (Note: number denoting total number of still births)	Number
History_4_Personal_Histroy			
1	Alcoholism_Yes/No	two values 1. Yes 2. No (Note: leave cell empty if its not mentioned in discharge summary explicitly)	Boolean
2	Alcoholism_Duration	Duration since start if yes	String
3	Tobacco_Use_Yes/No	two values 1. Yes 2. No (Note: leave cell empty if its not mentioned in discharge summary explicitly)	Boolean
4	Tobacco_Use_Duration	Duration since start if yes	String
5	Substance_Abuse_Yes/No	two values 1. Yes 2. No (Note: leave cell empty if its not mentioned in discharge summary explicitly)	Boolean
6	Substance_Abuse_Duration	Duration since start if yes	String

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7	Smoking_Yes/No	two values 1. Yes 2. No (Note: leave cell empty if its not mentioned in discharge summary explicitly)	Boolean
8	Smoking_Duration	Duration since start if yes	String
History_5_Family_Histroy			
1	Family History	Significant event or diseases in family	String
2	Standard Name	Description of the ICD 10 code	String - Optional Data
3	ICD 10	ICD 10 Code	String - Optional Data
4	SNOMED	SNOMED code	String - Optional Data
History_6_Past_History			
1	Past History	Acute disorder patient suffered in past or some procedure done in past should be recorded	String
2	Standard Name	Description of the ICD 10 code (CM for Diagnosis and PCS for Procedure)	String - Optional Data
3	ICD 10 Code	ICD 10 Code	String - Optional Data
4	SNOMED	SNOMED code	String - Optional Data
5	Duration	how much time has passed since event happened	String
Investigations			
1	Test Date	Date of test conducted	Date & Time
2	Parameter/Test Name	Name of test	String
3	Test Code	Hospital Specific Standard Code used by HIS for Test	String

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4	Test Category	Only to be selected out of following 15 1. Blood 2. Urine 3. CSF 4. Other Body Fluids 5. Imaging - USG 6. Imaging - X-Ray 7. Imaging - CT 8. Imaging - MRI 9. Imaging - PET 10. Imaging - Angiography 11. Endoscopy 12. Biopsy 13. Other Pathological Investigations 14. Other Radiological Investigation 15. Other Microbiological Investigation	String
4	Test Result	Result of the test	String
5	Test Range	Reference range of the test	String
6	Unit	Unit of the test results	String
Treatment_Details			
Treatment_Details_1_Procedures			
1	Type of Treatment	This field is standardised and there are the following values to be populated 1. Medical : If the patient was treated with Medical / Conservative management only 2. Intensive care : If patient was treated primarily in intensive care setup 3. Non-Allopathic treatment : If the patient was treated primarily with Non Allopathic treatment i.e., Ayurvedic, Unani, Occupational Therapy etc. 4. Surgical : If primary treatment is a surgical procedure 5. Investigation : If a patient is admitted for diagnostic interventions like colonoscopy, ugiscopy etc.	String

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2	Procedure ID	Hospital Specific ID for Procedure	String
3	Procedure Name	Name of the Procedure as given in discharge summary document (Note: Only to be filled if Surgical management)	String
4	Standard Procedure Name	ICD 10 PCS Code Description	String - Optional Data
5	SNOMED Code	SNOMED code	String - Optional Data
6	PCS Code	ICD 10 PCS Code	String - Optional Data
Treatment_Details_2_Medicines			
1	Medicine Name	Name of medicine given during hospitalisation	String
3	Generic Name	Salt composition	String
4	Frequency	Frequency of medication	String
5	Duration	Till how long medication was given	String
Post_Discharge			
Post_Discharge_1_Discharge_Data			
1	Discharge Type_Extracted Entity	Condition at discharge	String
2	Discharge_Type	There are 4 option to select from 1. Hemodynamically stable & fit for discharge 2. Referred for further management 3. Discharged against medical advice 4. Death	String
3	Diet Instruction	Diet Instruction	String
4	Follow up Date	Follow up Date	String
5	Followup Department Name	Followup Department Name-In case of multiple doctors and multiple department rows needs to be added accordingly	String

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6	Followup Doctor Name	Followup Doctor Name-In case of multiple doctors and multiple department rows needs to be added accordingly	String
Post_Discharge_2_Medicine			
1	Medicine Name	Name of medicine given during hospitalisation	String
3	Generic Name	Salt composition	String
4	Frequency	Frequency of medication	String
5	Duration	Till how long medication was given	String
Post_Discharge_3_Warning_Signs			
1	Warning Signs	Warning Signs	String
2	Standard Name	Description of the ICD 10 code	String - Optional Data
3	ICD 10	ICD 10 Code	String - Optional Data
4	SNOMED Code	SNOMED code of the Complaint name	String - Optional Data

IHX Payer Master

There are two types of mapping available :

- TPA based (Payer and Insurer ID)
- Non-TPA Insurer IDs

Scenario 1: Where the claim is submitted to the Processor, i.e., TPA (MediAssist, Volo, Vidal, Medsave, etc.) and the Payer is a different entity, i.e., HDFC, ABHI, Niva Bupa, etc. Here, the Processor is not the same as the Payer.

- payerID key value will refer to the 'Processor' (TPA): The one who processes the claim.
- insurerID key value will refer to the 'Payer' (insurer): The one who pays the money.
- Example: "payerID" = 523207 & "insurerID" = 30

Scenario 2: Where the Processor is the same as the Payer.

- payerID = insurerID
- HIS needs to send both the key values, payerID and insurerID, which will remain the same
- Example: "payerID" = 516571 & "insurerID" = 516571

All the payer information with payer names and paye ids are mentioned below for the reference of the HIS team

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PayerID	PayerName
516571	Aditya Birla Health Insurance Co. Limited
516572	Medi Assist Insurance TPA India Pvt Ltd
522771	MD India Healthcare (TPA) Services (Pvt.) Ltd.
523192	Heritage Health Insurance TPA Private Limited
523194	Genins India Insurance TPA Limited
523195	Health India Insurance TPA Services Private Limited
523196	Safeway Insurance TPA Private Limited
523197	Health Insurance TPA of India Limited
523207	MedSave Health Insurance TPA Limited
523375	GOOD HEALTH INSURANCE TPA LIMITED
523376	Universal Sompo General Insurance Co Ltd
523377	PARK MEDICLAIM TPA PVT LTD
523378	Ericson Insurance TPA Pvt. Ltd
523677	Raksha Health Insurance TPA Pvt Ltd
523678	Vidal Health Insurance TPA Private Limited
523679	Paramount Health Services & Insurance TPA Private Limited
523680	ICICI Lombard General Insurance
523681	Family Health Plan Insurance TPA Limited
523843	Bajaj Allianz General Insurance
523844	HDFC ERGO General Insurance
523845	Niva Bupa Health Insurance
523846	Volo Health Insurance TPA Pvt Limited(East West)
523847	Future Generali India Insurance Company (Corporate)
523848	IFFCO-TOKIO General Insurance Co. Limited
523849	Liberty General Insurance Limited
523850	Reliance General Insurance Co. Ltd
523893	Cholamandalam MS General Insurance
523894	Star Health and Allied Insurance Co Ltd
524231	Care Health Insurance
524232	United Healthcare Parekh TPA Private Limited
524800	Medvantage Insurance TPA Private Limited
526981	Tata AIG General Insurance

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526989	SBI General Insurance

Auth Package Automation Attributes

Package Procedure	Additional Mandatory Fields by Procedure	Expected Values	Limitation
Cataract	Eye	- Left - Right	NA
	Approach	- Conventional - MICS - SICS - Laser	NA
Normal Delivery	GPLA	Numeric	Limit: 30 characters
	Expected Date of delivery	- Date	NA
	Admission Type	- Planned - Emergency	NA
C-Section	GPLA	- Numeric	Limit: 30 characters
	Expected Date of delivery	- Date	NA
	Admission Type	- Planned - Emergency	NA
Dialysis	No. of dialysis sessions claimed	- Numeric	Between 1-20
Hysterectomy	Approach	- Open - Laparoscopic	NA
	Position	- Vaginal - Abdominal	NA
	Admission Type	- Planned - Emergency	NA
Adenotonsillectomy	Approach	- Conventional - Coblation	NA
Haemorrhoidectomy	Approach	- Conventional - Stapled - Laser	NA
Transurethral Prostatectomy (TURP)	No additional fields	NA	NA
Appendicectomy	Admission Type	- Planned - Emergency	NA

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Cholecystectomy	Admission Type	- Planned - Emergency	NA
URS with DJ stenting	Position	- Left - Right	NA
	Admission Type	- Planned - Emergency	NA

Claim Status Master for HIS Sync

Insurer Stage	Hospital Stage	IHX Status	IHX Status ID
Pre Authorization	New Admission	Pre-Auth Initiated	1
		Pre-Auth In Progress	31
		Pre-Auth Processing	26
		Pre-Auth Submitted to Payer	27
		Pre-Auth Approved	2
		Pre-Auth Query	5
		Pre-Auth Query Replied	6
		Pre-Auth Denied	7
	Enhancement	Enhancement Initiated	3
		Enhancement Approved	4
		Enhancement Query	40
		Enhancement Query Replied	41
		Enhancement Request Rejected	38
		Enhancement Denied	9
	Discharge	Discharge Initiated	8
		Discharge Approved	37
		Discharge Query	42
		Discharge Query Replied	43

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		Discharge Request Rejected	39
		Discharge Denied	61
	Early Discharge (Applicable only for HDFC claims)	Early Discharge Initiated	54
		Early Discharge Approved	55
		Early Discharge Denied	56
		Early Discharge Query	57
		Early Discharge Query Replied	58
Claim	Settlement	Settlement Initiated	10
		Claim in Progress	14
		Claim Approved	15
		Claim Denied	16
		Claim Query	17
		Claim Query Replied	52
		Settled	12
Common Status	Any	Processing	11
		Reconsider Request Submitted	53
		Reconsider Denied	62
		Cancelled	13

Response Format and dataTypes for Forward flows(IHX001 to IHX005)

API 1 : Get Patient Visit Details —> Sample Response JSON for Forward flows (IHX001 to IHX005) when there is a matching patient in HIS system.


```
{
 "hisData": {
 "hospitalId": "IHCU0015",
 "patientDetails": {
 "patientName": "SWARNENDU CHOWDHURY",
 "mobileNumber": "9804129083",
```


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```
"memberId": "MEMB123",
"emailId": "swarnendu@gmail.com",
"abhaId": "91231641712303",
"employeeId": "100",
"insurerName": "Aditya Birla",
"payer": "Medi Assist Insurance TPA India Pvt Ltd",
"payerID": "516572",
"insurerID": "28",
"corporateName": "Aditya Birla Health Insurance Co. Limited",
"policyNumber": "285620119034110",
"inpatientNumber": "298",
"UHID": "UHID001",
"policyType": "Retail",
"insuranceCardNumber": "!235408",
"aadharNumber": "234534219800",
"relationship": "Son",
"panCardNo": "JI091010Q",
"patientSignature": {
"fileName": "swarnendu.png",
"fileReferenceId": "https://customer-data.s3.us-east-1.amazonaws.com/
documents/report-2025.pdf?X☑Amz-Algo rithm=AWS4☑HMAC☑SHA256&X☑Am
z-Credential=AKIAIOSFODNN7EXAMPLE%2F20250421 %2Fus-east-1%2Fs
3%2Faws4_request&X☑Amz-Date=20250421T123456Z&X☑Amz-Expires= 360
0&X☑Amz-SignedHeaders=host&X☑Amz-Signature=9d80c48b952596c6738b
b5d932cbfb6 48d5ea422dfbd86f421d6dd30837c4c78"
}
},
"treatmentDetails": {
"preAuthType": "Dialysis",
"reasonForHospitalization": "Ailments",
"systemOfMedicine": "Allopathy",
"numberOfDialysis"↵ 2,
"routeOfAdministration": "Oral",
"sinceWhen": "80",
"sinceWhenUnit": "Days",
"surgeryType": "Major",
"dateOfSurgery": "1730222721",
```


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```
"conditionOnDischarge": "Alive",
"admissionType": "Planned",
"doctorQualification": "MBBS",
"speciality": "Critical Care",
"stateMedicalCouncil": "Karnataka",
"presentComplaints": "free text",
"isPatientDischargeToday": false,
"isCovidCase": false,
"eye": "right",
"approach": "Conventional",
"dateOfAdmission": "1730222721",
"dateOfDischarge": "1730226660",
"proposedLineOfTreatment": {
"id"↵ 1112,
"name": "Medical Management"
},
"automatedProcedure": {
"id"↵ 16,
"procedure": "Cataract",
"proposedLineOfTreatmentID"↵ 1112
},
"procedure": [
{
"id"↵ 2347,
"name": "Abdominal hysterectomy"
},
{
"id"↵ 276,
"name": "Repair Right Knee Bursa and Ligament, Open Approach"
}
],
"diagnosis": [
{
"icdCode": "A90",
"name": "Dengue fever [classical dengue]"
},
{
```


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```
"icdCode": "Q12.0",
"name": "Congenital cataract"
}
]
},
"doctorDetails": {
"doctorName": "John Doe",
"doctorRegistrationNo": "DR174",
"roomType": "Economy Room",
"isDischargedToday": true,
"isDeath": false,
"doctorSpeciality": {
"id"↵ 241635,
"name": "Addiction psychiatrist"
},
"doctorSignature": {
"fileName": "JohnDoe.png",
"fileReferenceId": "https://acme-corp.s3.eu-west-1.amazonaws.com/fina
ncial/annual-report-2024.xlsx?X☑Amz-Al gorithm=AWS4☑HMAC☑SHA256&X☑A
mz-Credential=AKIAEXAMPLE123456789%2F2025042 1%2Feu-west-1%2Fs
3%2Faws4_request&X☑Amz-Date=20250421T140523Z&X☑Amz-Expires=7200
&X☑Amz-SignedHeaders=host&X☑Amz-Signature=b31df9a8e417bdb9c957d7b
35441f8 df2d92e6c88735a5db64bd82a3a0f8ce3a"
}
},
"treatmentPastHistory": {
"pastAilments": [
{
"id": "242715",
"value": "External prolapsed hemorrhoids",
"durationYear"↵ 1,
"durationMonth"↵ 6
},
{
"id": "242726",
"value": "Bronchiectasis",
"durationYear"↵ 1,
```


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```
"durationMonth"↵ 6
}
],
"otherAilments": "Dimentia"
},
"caseDetails": {
"dateOfInjury": "1730222721",
"firNumber": "FIR1234",
"isRTA": false,
"isSubstanceAbuse": false,
"isReportToPolice": false,
"isTestConducted": false,
"injuryCausedBy": "Accident",
"accidentDueTo": "Reason",
"isMLC": false
},
"billDetails": {
"package"↵ 0,
"infraCharges"↵ 1000,
"consultation"↵ 2000,
"otCharges"↵ 3500,
"surgeon"↵ 5000,
"laboratoryInvestigation"↵ 500,
"pharmacy"↵ 1000,
"otherExpenses"↵ 0,
"medicine"↵ 2000
},
"dischargeInitiationDetails": {
"isUploadSettlementDocument": true,
"dateOfAdmission": "1730222721",
"dateOfDischarge": "1730226660",
"paUtilised": false,
"paidAmount"↵ 1400,
"inpatientNumber": "69515",
"receiptNumber": "IPSR54153",
"visitType": "IP"
```


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```
},
"documentsMetaData": [
{
"fileReferenceId": "https://customer-data.s3.us-east-1.amazonaws.com/
documents/report-2025.pdf?X✍Amz-Algo rithm=AWS4✍HMAC✍SHA256&X✍Am
z-Credential=AKIAIOSFODNN7EXAMPLE%2F20250421 %2Fus-east-1%2Fs
3%2Faws4_request&X✍Amz-Date=20250421T123456Z&X✍Amz-Expires= 360
0&X✍Amz-SignedHeaders=host&X✍Amz-Signature=9d80c48b952596c6738b
b5d932cbfb6 48d5ea422dfbd86f421d6dd30837c4c78",
"fileName": "fileName1.pdf",
"uploadedOn": "1622865600000",
"docTypeID"↵ 148,
"docTypeName": "Bills"
},
{
"fileReferenceId": "https://acme-corp.s3.eu-west-1.amazonaws.com/fina
ncial/annual-report-2024.xlsx?X✍Amz-Al gorithm=AWS4✍HMAC✍SHA256&X✍A
mz-Credential=AKIAEXAMPLE123456789%2F2025042 1%2Feu-west-1%2Fs
3%2Faws4_request&X✍Amz-Date=20250421T140523Z&X✍Amz-Expires=7200
&X✍Amz-SignedHeaders=host&X✍Amz-Signature=b31df9a8e417bdb9c957d7b
35441f8 df2d92e6c88735a5db64bd82a3a0f8ce3a",
"fileName": "fileName2.pdf",
"uploadedOn": "1622865600000",
"docTypeID"↵ 147,
"docTypeName": "ClaimForm"
},
{
"fileReferenceId": "https://user-uploads.s3.ap-southeast-2.amazonaws.c
om/images/profile-pic-1234.jpg?X✍Amz Algorithm=AWS4✍HMAC✍SHA256&X✍
Amz-Credential=AKIAJEXAMPLEK3Y%2F20250421%2 Fap-southeast-2%2Fs
3%2Faws4_request&X✍Amz-Date=20250421T151030Z&X✍Amz-Expire s=900
&X✍Amz-SignedHeaders=host&X✍Amz-Signature=5c82f19d702a9e10b57e9a2
d7f6d248 a34b3c4f2be8c7f294e54cc8a013e6f8d",
"fileName": "fileName3.pdf",
"uploadedOn": "1622865600000",
"docTypeID"↵ 150,
```


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```
"docTypeName": "ConsultationPapers"
},
{
"fileReferenceId": "https://media-archive.s3.us-west-2.amazonaws.com/
videos/product-demo-v2.mp4?X☑Amz-Al gorithm=AWS4☑HMAC☑SHA256&X☑
Amz-Credential=AKIAEXAMPLE987DEMO%2F20250421 %2Fus-west-2%2Fs
3%2Faws4_request&X☑Amz-Date=20250421T160045Z&X☑Amz-Expires= 432
KIMS 🖩 IHX
6
00&X☑Amz-SignedHeaders=host&X☑Amz-Signature=f3a7d88e7ae4e2cf6ec3a
785d4e3f2f9 e8b72c6d731b4f82dc1f8d6c5a8c17de",
"fileName": "fileName4.pdf",
"uploadedOn": "1622865600000",
"docTypeID"↵ 149,
"docTypeName": "DischargeSummary"
},
{
"fileReferenceId": "https://media-archive.s3.us-west-2.amazonaws.com/
videos/product-demo-v2.mp4?X☑Amz-Al gorithm=AWS4☑HMAC☑SHA256&X☑
Amz-Credential=AKIAEXAMPLE987DEMO%2F20250421 %2Fus-west-2%2Fs
3%2Faws4_request&X☑Amz-Date=20250421T160045Z&X☑Amz-Expires= 432
00&X☑Amz-SignedHeaders=host&X☑Amz-Signature=f3a7d88e7ae4e2cf6ec3a
785d4e3f2f9 e8b72c6d731b4f82dc1f8d6c5a8c17de",
"fileName": "fileName5.pdf",
"uploadedOn": "1622865600000",
"docTypeID"↵ 152,
"docTypeName": "KYC"
}
]
},
"isSuccess": true,
"errorMessage": null,
"transactionId": "1a353efd-01f4☑45ac-8045-da33dbda292b
}
```


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API 2 : Document Fetch

Sample request body:

```
{
    "hospitalId": "1234",
    "uhid": "UHID123"
    "ipNumber": "IP123",
    "fileName": "DischargeSummary.pdf",
    "hisFileIdentifier": "{unique file identifier}"
}
```

Sample Response:


```
{
    "isSuccess": true,
    "errorMessage": null,
    "response": {
        "ipNumber": "IP123",
        "uhid": "UHID123",
        "hospitalId": "1234",
        "letters": [
            {
                "fileName": "DischargeSummary.pdf",
                "letterStream":
"JVBERi0xLjQKJeLjz9MKMiAwIG9iago8PC9MZW5ndGggMTAvRmlsdGVyL0ZsYXRlRGVjb2RlPj5zd
HJlYW0KeJwr5AIAAO4AfAplbmRzdHJlYW0KZW5kb2JqCjMgMCBvYmoKPDwvTGVuZ3RoIDMxNzUvRml
sdGVyL0ZsYXRlRGVjb2RlPj5zdHJlYW0KeJztXUtv5LgRvvev0DmAZZGUSAlYLDDTtoPkEGAwBvawy
CGY2dnFwt7E2UP+fqhHS1R9ooqiJbU9kAcYW9USq1isN4vq279+/lfy65/J7fnzf5Iv3e/z51OWZjp
rf5L6340LKHSqq+Yn0VKlRjbQL8+nl+Tl9On0yf5f/345CZkWziDk0j5wQdsC/vzyx+m2JejUQj6f/
2H/+l8ik7/bq9+Tn/9pf33tcNQ3PJ/KTOapJcXUl0/uZWlUkea5VNLC4bK++bfTT39J/qgJtdPL7Fz
            }
        ]
    }
}
```


API 3 : Callback Flow

Sample request body:

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```
{
 "hospitalId": "IHCU0015",
 "ihxReferenceID"↵ 2089010,
 "hospActionFlowID"↵ 1080010091,
 "claimNumber": "39361779",
"uhid": "UHID001",
 "ipNumber": "298",
 "mobileNumber": "8167486541",
 "abhaAddress": "63095836365693@sbx",
 "abhaNumber": "99999999999999",
 "statusId"↵ 1,
 "statusName": "Pre-Auth Approved",
 "claimedAmount"↵ 90000,
 "approvedAmount"↵ 84126,
 "settledAmount"↵ 84126,
 "tdsAmount"↵ 100,
 "utrNo": "UTR12345",
 "letters": [
 {
 "filename": "PayerFile.pdf",
 "letterStream": ""
 },
 {
 "filename": "PAApproved.pdf",
 "letterStream": ""
 }
 ],
 "remarks": "Pre auth approved received from payer"
 }
```


Sample response


```
{
"isSuccess": true,
 "errorMessage": null,
 "response": "Record Saved Successfully",
 "transactionId": "${a unique id}"
```


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```
}
```

Error Response Handling

FORWARD FLOW - IHX001 to IHX005

Scenario	Expected HTTP Response Code	Expected mandatory fields in response body

The HIS system is down or un reachable or any other unhandled error from HIS system	5XX	If application/code breaks and handled in code,
		
		{
		isSuccess: false,
		errorMessage: “Something went wrong, please try after sometime”,
		response: null,
		identifier: “${a unique id}”
		}

No matching patient in HIS system	200 OK	{
		isSuccess: false,
		errorMessage: “Patient Not found”,
		response: null,
		identifier: “${a unique id}”
		}

Any validation errors during calling HIS APIs.	400 Bad Request	{
		isSuccess: false,
		errorMessage: “whatever the validation error message is”,
		response: null,
		identifier: “${a unique id}”
		}

There is a matching patient in HIS system	200 OK	{
		"isSuccess": true,
		"errorMessage": null,
		"identifier": "${a unique id}",
		"response": {
		"hospitalId": "JL001",
		"patientDetails": {
		"patientName": "John Doe",
		"mobileNumber": "9809878760",
		"memberId": "MEMB5679890",
		"emailId": "johndoe@gmail.com",
		"abhaId": "56787898791230",
		"employeeId": "EMP78680",
		"insurerName": "Medi Assist",
		"payer": "Niva Bupa",

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		"payerID": "532902",
		"insurerID": "560301",
		"corporateName": "FAMILY HEALTH PLAN LIMITED",
		"policyNumber": "POLC8798008790",
		"inpatientNumber": "IP1234",
		"registrationNumber": "REG345678901234"
		},
		"treatmentDetails": {
		"dateOfAdmission": "1622865600000",
		"dateOfDischarge": "1623547261000",
		"lineOfTreatment": "Surgical Management",
		"diagnosis": "ICD-10 code of diagnosis"
		},
		"doctorDetails": {
		"doctorName": "Dr. Jane Smith",
		"doctorRegistrationNo": "DOC67980",
		"roomType": "ICU 02",
		"isDischargedToday": false,
		"isDeath": false
		},
		"treatmentPastHistory": {
		"pastAilments": "HyperTension",
		"pastAilmentDuration": "2",
		"otherAilments": "Headache"
		},
		"caseDetails": {
		"dateOfInjury": "1622865600000",
		"firNumber": "FIR7890",
		"isRTA": true,
		"isSubstanceAbuse": false,
		"isReportToPolice": true,
		"isTestConducted": true
		},
		"billDetails": {
		"package": 50000,
		"infraCharges": 10000,
		"consultation": 1000,
		"otCharges": 500,
		"surgeon": 25000,
		"laboratoryInvestigation": 3000,
		"pharmacy": 5000,
		"otherExpenses": 1000
		},
		"dischargeInitiationDetails": {
		"dateOfAdmission": "1622865600000",
		"dateOfDischarge": "1623547261000",
		"paUtilised": true,
		"paidAmount": 1000,
		"inpatientNumber": "IP1234",
		"receiptNumber": "RECP1234"

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		}, "letters": [{ "fileName": "AuthorizationLetter.pdf", "uploadedOn": "1622865600000", "letterStream": "JVBERi0xLjcKJeLjz9MKNCAwIG9iago8PAovVHlwZSAvWE9iamVjdAovU3VidHlwZSAvSW1hZ2UKL1dpZHRoIDEKL0hlaWdodCAxCi9CaXRzUGVyQ29tcG9uZW50IDgKL0NvbG9yU3BhY2UgL0RldmljZVJHQgovTGVuZ3RoIDMKPj4Kc3RyZWFtCv///wplbmRzdHJlYW0KZW5kb2JqCjUgMCBvYmoKPDwKL0ZpbHRlciAvRmxhdGVEZWNvZGUKL0xlbmd0aCA0NQo+PgpzdHJlYW0KeJwr5DK1NFUwAEJjYxM9MwtzIMvI1FjPzNRMITmXSz/CQMElnyuQCwChPggOCmVuZHN0cmVhbQplbmRvYmoKMyAwIG9iago8PAovVHlwZSAvUGFnZQovTWVkaWFCb3ggWzAgMCA1OTUgODQyXQovUmVzb3VyY2VzIDw8Ci9YT2JqZWN0IDw8Ci9YMCA0IDAgUgo+Pgo+PgovQ29udGVudHMgNSAwIFIKL1BhcmVudCAyIDAgUgo+PgplbmRvYmoKNCAwIG9iago8PAovTGVuZ3RoIDYzMzwMjcwOTQ4NDRaKQo+PgplbmRvYmoKNyAwIG9iago8PAovU2l6ZSA4Ci9Sb290IDEgMCBSCi9JbmZvIDYgMCBSCi9JRCBbPEEwODNGRDczRDg2OUMyMUE1N0E1NURGNjE2NjZDM0UwPiA8NEM4QkVBNTFGOUU2QzU1M0NDOUE0MkI4QTIyREVBRjQ+XQovVHlwZSAvWFJlZgovVyBbMSAyIDJdCi9GaWx0ZXIgL0ZsYXRlRGVjb2RlCi9JbmRleCBbMCA4XQovTGVuZ3RoIDM5Cj4+CnN0cmVhbQp4nGNgYPj/n5HxMSAyIDI/kGBkPEV

		BhcmVudCAyIDAgUgo+PgplbmRvYmoKMiAwIG9iago8PAovVHlwZSAvUGFnZXMKL0tpZHMgWzMgMCBSXQovQ291bnQgMQo+PgplbmRvYmoKMSAwIG9iago8PAovVHlwZSAvQ2F0YWxvZwovUGFnZXMgMiAwIFIKPj4KZW5kb2JqCjYgMCBvYmoKPDwKL1Byb2R1Y2VyIChpTG92ZVBERikKL01vZERhdGUgKEQ6MjAyMzEwMjcwOTQ4NDRaKQo+PgplbmRvYmoKNyAwIG9iago8PAovU2l6ZSA4Ci9Sb290IDEgMCBSCi9JbmZvIDYgMCBSCi9JRCBbPEEwODNGRDczRDg2OUMyMUE1N0E1NURGNjE2NjZDM0UwPiA8NEM4QkVBNTFGOUU2QzU1M0NDOUE0MkI4QTIyREVBRjQ+XQovVHlwZSAvWFJlZgovVyBbMSAyIDJdCi9GaWx0ZXIgL0ZsYXRlRGVjb2RlCi9JbmRleCBbMCA4XQovTGVuZ3RoIDM5Cj4+CnN0cmVhbQp4nGNgYPj/n5HxHAMDIDI/kZW5kc3RyZWFtCmVuZG9iagpzdGFydHhyZWYKNTgyCiUlRU9GCg== " }] } }

CALLBACK FLOW - IHX006

Scenario	Expected HTTP Response Code	Expected mandatory fields in response body

The HIS system is down or unreachable or any other unhandled error from HIS system	5XX	If application/code breaks and handled in code,
		
		{
		isSuccess: false,
		errorMessage: “Something went wrong, please try after sometime”,
		response: null,
		identifier: “${a unique id}”
		}

No matching patient in HIS system	200 OK	{
		isSuccess: false,
		errorMessage: “Patient Not found”,
		response: null,
		identifier: “${a unique id}”
		}

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Any validation errors during calling HIS APIs.	400 Bad Request	{
		isSuccess: false,
		errorMessage: “whatever the validation error message is”,
		response: null,
		identifier: “${a unique id}”
		}

There is a matching patient in HIS system and successfully saved the record	200 OK	{
		isSuccess: true,
		errorMessage: null,
		response: “Record Saved Successfully”,
		identifier: “${a unique id}”
		}

IHX Integration Assessment: Questionnaire for Workflow Understanding & Customization Scope

1. Are you looking for any customizations or changes in the suggested approach?
2. Is the HIS deployed on-premise or cloud-based?
3. What unique identifiers are used for patients (UHID, IP Number, ABHA, etc.)?
4. Can patient details be fetched in real-time based on any of the identifiers?
5. Do you need past visits to be fetched on IHX? Is it possible to fetch all past visits of a patient along with visit codes?
6. Does the HIS differentiate between OPD, IPD, Emergency, and Daycare admissions?
7. Does your HIS have fields related to insurance details like member id, employee id, policy no.,etc.?
8. Does your HIS support multiple insurers and its details against the same patient?
9. Is there a provision to handle base and top-up claims if IHX pushes back the information?
10. Does your system maintain a visit history or episode history per patient?Can you expose all previous admissions for the same UHID via API?
11. Is ICD-10 coding used for diagnosis? If not, how is diagnosis represented?
12. Are treatment plans captured structurally or free-text in HIS?
13. Is there a field for the proposed line of treatment (Surgical/Medical/ICU/Investigation)?
14. How is the admission type (planned/emergency) tracked?
15. Are doctor specializations stored in a master table?
16. Is the discharge process initiated manually or based on clinical workflows?
17. How is room allocation and room type tracked over the course of admission?
18. Can HIS expose dynamic room type changes across admission duration?
19. Is a hospital bill generated daily, or at discharge only?
20. Can itemized bills be exposed in JSON format?
21. Does the billing engine categorize charges as per IRDAI L1 structure?
22. How will documents shared with IHX be via base64 or FTP URLS?

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23. Can each billed item include unit price, quantity, tax, and discount details?
24. Are packages and non-package items billed in the same or different modules?
25. Is there a provision for package override and documentation thereof?
26. How are rejection letters or insurer queries stored and accessed?
27. Does HIS maintain a history of insurer communication (queries, approvals, etc.)?
28. How does your system receive and handle external status updates (e.g., claim approved)?
29. Where are UTR numbers and settlement amounts stored?
30. Is it possible to view query remarks or insurer comments in HIS?
31. Are callback APIs supported to update HIS with claim status?
32. What mechanism is used for version control of API fields and schema changes?
33. How are document uploads handled – via local uploads, scanners, or directly from HIS?
34. What file types and size limits does the HIS support for uploads?
35. Can you share patient documents like claim form, discharge summary, and reports via API?
36. How often is your test/staging environment refreshed to match production?
37. Is a data dictionary or metadata mapping available for your current HIS structure?
38. What is the level of flexibility in HIS to customize or extend data fields?
39. What are the current limitations of HIS that may affect seamless IHX integration?
40. How is claim form finalization coordinated across departments (billing, medical, insurance)?
41. Is the insurance desk trained on digital submissions, or is manual support required?
42. Is there an escalation matrix if claim data is incomplete or incorrect in HIS?
43. Is there a dedicated testing team or UAT support from the HIS vendor?
44. Does HIS support multi-tenancy or data isolation across hospitals if shared?
45. How do you ensure data security and authorization when sharing patient data externally?

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